

MODBUS-RTU Communication Protocol of the Meteo Station

Version: V1.0

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1.1 Communication Cable Description

(1) Communication port: 232; communication cable marking: COM232; and signal output manner: 232.

Communication pin definition:

4-core aviation connector	9-core serial port	Pin definition	Cable colour
2 pins	2 pins	Receiving signal (RX)	Green
3 pins	3 pins	Transmitting signal (TX)	Blue
4 pins	5 pins	Grounding (GND)	Black

(2) Communication port: 485; communication cable marking: 485 communication; and signal output manner: 485.

Communication pin definition:

4-core aviation connector	9-core serial port	Pin definition	Cable colour
2 pins	Connecting pin 2 and pin 4	485A	Green
3 pins	Connecting pin	485B	Blue

	1 and pin 3		
	Connecting pin 6 and pin 7		
	Connecting pin 8 and pin 9		

Remark: Applicable to devices of PC-4 series.

1.2 Baud Rate

The device communicates in two-way asynchronous communication manner. It is specified that there is 1 start bit, 8 data bits, 1 stop bit, and no parity check. Three baud rates: 4800, 9600, 19200 are supported. The baud rate of a device can be selected via the menu. Alternatively, the baud rate can be selected or changed via the command if the device is equipped with no LCD.

Default serial port communication parameters:

Serial port baud rate: 9600

Stop bit: 1

Data bit: 8

Check bit: Null

Default device address: 01

1.3 Instruction Format

Additional address	Function code	Data	Error check
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1.4 Setting of Additional Address and Communication Parameters

The additional address is single-byte hexadecimal data, and the device only responds to the request data frame whose additional address is consistent with the local additional address. The address can be set via the menu of the system software or the command.

1.5 Data

According to the Modbus protocol, the hexadecimal data and the floating-point data comply with the rule that high byte followed by low byte.

1.6 Function Code

Function code	Description
0x01	Read coils
0x05	Write coils
0x03	RDSR
0x10	Write register

1.7 Error Check

Cyclic redundancy check (CRC) is adopted, and the CRC result is single word, where low byte is followed by high byte. The formation law is as follows.

- Set a 16-bit CRC register and assign an initial value 0xFFFF to it.
- Perform bitwise exclusive-OR operation on the first byte of the data frame (the additional address) and the least significant 8 bits, and save the result into the CRC register.
- Shift the CRC register to the right by 1 bit and detect whether the least

significant bit is 1. If so, perform exclusive-OR operation on the CRC register and the fixed number 0xA001.

- d) Repeat step c for 8 times.
- e) Perform steps b, c, and d on the next byte of the data frame till the last data in the data field.
- f) The final content in the CRC register is the check value, and append the value to the last data of the data frame, where the least significant 8 bytes are followed by the most significant bytes.

1.8 Read Register (Access Real-time Monitoring Data)

Request frame	Byte size	Value
Additional address	1	0x01~0xFF
Function code	1	0x03
Start address	2	0x0000~0xFFFF
Register quantity	2	n=0x0001~0xFFFF
Check code	2	
Reply frame	Byte size	Value
Additional address	1	0x01~0xFF
Function code	1	0x03
Byte(s)	1	2 * n
Register value 1-n	2 * n	

Check code	2	
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Read sensor data register

For example, the device address is 01, to obtain the real-time monitoring data.

Send the request: 0x01 0x03 0x00 0x00 0x00 0x0A 0xC5 0xCD

Replay data:

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01 03 56 01 0c 00 7d 00 7d 00 7d 00 7d 00 7d 02 54 00
19 00 19 00 19 15 f7 00 0d 27 29 03 7c 03 7c 03 7c 03 7c
03 7c 03 7c 01 0e 00 17 00 70 00 68 00 66 00 05 01 70 01 70 01
70 01 70 01 70 01 70 01 70 00 7e 00 b4 47 c1 47 c1 47 c1 47 c1
47 c1 47 c1 47 c1 4CA0
```

1.9 Register Address

Address	Item	Resolution	Example data	Sample data analysis	Signed number	Standby
0000	Ambient temperature	0.1	0x010c	26.8°C	√	
0001	Temperature 1 (Bracket)	0.1	0x007d	12.5°C	√	
0002	Temperature 2	0.1	0x007d	12.5°C	√	√
0003	Temperature 3	0.1	0x007d	12.5°C	√	√
0004	Temperature 4	0.1	0x007d	12.5°C	√	√
0005	Temperature 5	0.1	0x007d	12.5°C	√	√
0006	Dew-point temperature	0.01	0x007d	1.25°C	√	√
0007	Ambient humidity	0.1	0x0254	59.6%RH		
0008	Soil moisture 1	0.1	0x0019	2.5m3/ m3	√	
0009	Soil moisture 2	0.1	0x0019	2.5m3/ m3	√	
000A	Soil moisture 3	0.1	0x0019	2.5m3/ m3	√	
000B	CO ₂	0.1	0x15 f7	562.3PPm	√	

000C	Evaporation	0.1	0x000d	1.3mm	√	
000D	Air pressure	0.1	0X2729	1002.5HPa		
000E	Instantaneous value of total radiation 1	1	0x037c	892 W/ m ²		
000F	Instantaneous value of scattered radiation	1	0x037c	892 W/ m ²		
0010	Instantaneous value of direct radiation	1	0x037c	892 W/ m ²		
0011	Instantaneous value of total radiation 2	1	0x037c	892 W/ m ²		
0012	Instantaneous value of net radiation	1	0x037c	892 W/ m ²	√	
0013	Instantaneous value of photosynthetic radiation	1	0x037c	892 W/ m ²	√	
0014	Instantaneous value of ultraviolet radiation	1	0x037c	892 W/ m ²	√	
0015	Instantaneous wind direction	1	0x010e	270°		
0016	Instantaneous wind speed	0.1	0x0017	2.3 m/s		
0017	Wind speed in 2 minutes	0.1	0x0070	11.2 m/s		
0018	Wind speed in 10 minutes	0.1	0x0068	10.4 m/s		
0019	Cumulative value of rainfall	0.1	0x0066	10.2mm	√	
001A	Cumulative value of sunshine duration	1	0x0005	5 min	√	
001B	Cumulative value of total radiation 1	0.001	0x0170	0.368MJ/ m ²		
001C	Cumulative value of scattered radiation	0.001	0x0170	0.368MJ/ m ²		
001D	Cumulative value of direct radiation	0.001	0x0170	0.368MJ/ m ²		
001E	Cumulative value of total radiation 2	0.001	0x0170	0.368MJ/ m ²		
001F	Cumulative value of net radiation	0.001	0x0170	0.368MJ/ m ²		√
0020	Cumulative value of photosynthetic	0.001	0x0170	0.368MJ/ m ²		√

	radiation					
0021	Cumulative value of ultraviolet radiation	0.001	0x0170	0.368MJ/ m ²		√
0022	Daily rainfall	0.1	0x007e	12.6mm		√
0023	Daily sunshine duration	1	0x00b4	180 min		√
0024	Daily cumulative value of total radiation 1	0.001	0x47c1	18.369MJ/ m ²		
0025	Daily cumulative value of scattered radiation	0.001	0x47c1	18.369MJ/ m ²		
0026	Daily cumulative value of direct radiation	0.001	0x47c1	18.369MJ/ m ²		
0027	Daily cumulative value of total radiation 2	0.001	0x47c1	18.369MJ/ m ²		
0028	Daily cumulative value of net radiation	0.001	0x47c1	18.369MJ/ m ²		√
0029	Daily cumulative value of photosynthetic radiation	0.001	0x47c1	18.369MJ/ m ²		√
002A	Daily cumulative value of ultraviolet radiation	0.001	0x47c1	18.369MJ/ m ²		√

Note:

When one radiation sensor is connected on site:

(1) If the sensor is installed horizontally, the total radiation 1 is radiation on the horizontal surface.

(2) If the sensor is installed in a slant manner, the total radiation 1 is radiation on the slant surface.

When two radiation sensors are connected on site:

(1) Sensor 1 records the radiation on slant surface, and total radiation 1 is the radiation on the slant surface.

(2) Sensor 2 records the radiation on horizontal surface, and total radiation 2 is the radiation on the horizontal surface.

Appendix

1.1 Setting Baud Rate Instruction

In setting the baud rate instruction, basic communication parameters do not need to be changed, but the protocol format and sending manner are different from those of the Modbus protocol. The sending manner is ASCII code, and the packet format is as follows:

Sending the packet:

#Additional address 210B, baud rate GG (the baud rate is hexadecimal data)

Receiving the packet:

#Additional address 210BGG

For example:

The additional address is 01, and the baud rate is set to 9600

Sending: #01210B2580GG

Receiving: #01210BGG

1.2 Setting Additional Address Instruction

In setting the additional address instruction, basic communication parameters do not need to be changed, but the protocol format and sending manner are different from those of the Modbus protocol. The sending manner is ASCII code, and the packet format is as follows:

Sending the packet:

#1400 additional address GG (the additional address ranges from 01 to FF)

Receiving the packet:

#Additional address 140OGG

For example:

Setting the additional address to 01

Sending: #140O01GG

Receiving: #01140OGG